

Group-MoE: Learned Dispatch to Algebraic Fixed-Function Units

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Abstract

Neural networks that encounter symmetry either enforce it rigidly in every layer or ignore it entirely. We propose GROUP-MOE, a mixture-of-experts architecture where each expert implements a finite group representation in the irreducible representation (irrep) basis, and a learned router detects which symmetry applies. This gives models *algebraic fixed-function units*—analogous to dedicated hardware accelerators—that the network learns to invoke selectively. We introduce the *complement split*, a controlled evaluation methodology that isolates symmetry transfer from memorization, and validate the architecture on synthetic tasks. Our experiments demonstrate: (1) complement transfer from seen to unseen orderings (+16 percentage points for S_2 , +3.9pp for S_3 beyond a matched standard MoE), (2) near-perfect zero-shot compositional generalization from transpositions to 3-cycles via irrep composition, (3) correct router discrimination between non-nested groups, and (4) zero-degradation compatibility as a transformer FFN replacement. We analyze when algebraic structure outperforms lookup-table memorization and identify the combinatorial scaling regime where group experts should provide their greatest advantage.

1 Introduction

Symmetry is ubiquitous in the data that neural networks process. Images have rotational and reflective symmetries. Molecules have point-group symmetries. Language has permutation symmetries: “Alice likes Bob” and “Bob is liked by Alice” describe the same relationship regardless of entity ordering. The question is not whether symmetry exists, but how a model should handle it.

The field has converged on two approaches, each with significant limitations. *Geometric deep learning* [1, 2] builds equivariance into every layer, ensuring that symmetry transformations of the input produce corresponding transformations of the output. This is correct by construction but rigid: the model cannot mix symmetric and non-symmetric computation, and applying the wrong symmetry is as harmful as applying none. *Standard architectures* (MLPs, transformers) ignore symmetry entirely and hope the right behavior emerges from training data. Our prior work [11] showed that large language models do achieve *functional* equivariance—correct outputs under input permutation—but without developing *structural* equivariance—group representations in their latent space. They memorize the multiplication table rather than learning algebra.

This paper proposes a third path: **group representations as optional expert modules** with learned dispatch. The analogy to hardware design is precise. Just as processors evolved from general-purpose CPUs to include fixed-function accelerators (FFT units, matrix multiply engines, cryptographic co-processors), neural architectures can include *algebraic fixed-function units*—modules hardwired in the irreducible representation basis of a finite group—that are activated only when the router detects the corresponding symmetry. The irrep basis provides the same kind of structural compression that fixed-function hardware provides over general-purpose computation: a $k \times k$ block-diagonal transform instead of a $d \times d$ dense matrix, with exact composition guaranteed by the group homomorphism property.

Our contributions:

1. The GROUP-MOE architecture: group representations in the irrep basis as expert modules, combined with a learned symmetry router and residual blending (Section 2).
2. The *complement split*: a controlled evaluation methodology that isolates symmetry transfer from memorization by ensuring every test example requires generalization from a seen ordering to an unseen one (Section 3).

3. Experimental validation on five synthetic tasks, each answering one question about the architecture (Section 4).
4. Analysis of when algebraic structure outperforms lookup-table memorization, connecting our findings to the broader question of scaling (Section 5).

2 Architecture

2.1 Group Expert Module

The core building block of GROUP-MOE is the *group expert*, which applies a group representation in a learned subspace of the activation space. For a finite group G with irreducible representations of total dimension k , the expert implements:

$$x \xrightarrow{P} z \xrightarrow{R(g)} z' \xrightarrow{P^\dagger} x' \quad (1)$$

where $x \in \mathbb{R}^d$ is the input activation, $P \in \mathbb{R}^{k \times d}$ is a learned projection into the symmetry-active subspace, $R(g) \in \mathbb{R}^{k \times k}$ is the irrep matrix for group element g (fixed, not learned), and $P^\dagger \in \mathbb{R}^{d \times k}$ is the injection back to the full space.

The irrep matrix $R(g)$ is block-diagonal by construction:

$$R(g) = \begin{pmatrix} \rho_1(g) & & \\ & \rho_2(g) & \\ & & \ddots \end{pmatrix} \quad (2)$$

where each ρ_i is an irreducible representation. For S_3 , the symmetric group on three elements, the irreps are: trivial (1D), sign (1D), and standard (2D), giving $k = 4$. For a model with $d = 128$, this provides $32 \times$ compression relative to a full $d \times d$ matrix.

The critical property is **exact composition**: $R(g_1)R(g_2) = R(g_1g_2)$ by the group homomorphism. This means the expert’s behavior on composed group elements is determined entirely by its behavior on generators—a property no learned transform can guarantee.

2.2 Symmetry Router

The router is a lightweight MLP that maps each activation to a distribution over routing options: one pass-through option plus one option per group element across all available groups. For a model with groups $G_1, \dots, G_m$:

$$\text{options} = \{\text{pass-through}\} \cup \bigcup_{i=1}^m \{g \mid g \in G_i\} \quad (3)$$

The router uses hard routing (argmax) with soft confidence blending:

$$x' = x + \alpha \cdot (\text{expert}(x, g) - x) \quad (4)$$

where α is the router’s confidence score for the selected option. Pass-through tokens ($\alpha = 0$ or pass-through selected) pass unchanged, ensuring that symmetry is only applied when detected.

2.3 Load Balancing

Following the Switch Transformer [5], we add an auxiliary load-balancing loss to prevent expert collapse:

$$\mathcal{L}_{\text{balance}} = N \sum_{i=1}^N f_i \cdot p_i \quad (5)$$

where N is the number of routing options, f_i is the fraction of tokens assigned to option i (hard), and p_i is the mean routing probability for option i (soft). This encourages uniform utilization of all options without forcing specific routing patterns.

2.4 Integration

The GROUP-MOE layer is a residual module that handles both (B, d) and (B, L, d) inputs by flattening the sequence dimension internally. This makes it a drop-in replacement for any feedforward block in a transformer or MLP architecture.

Table 1 lists the implemented groups with their irrep dimensions and parameter counts.

Table 1: Implemented group representations.

Group	Order	Irreps	k	Expert params ($d=128$)
$\mathbb{Z}_2$	2	trivial + sign	2	512
$\mathbb{Z}_3$	3	trivial + standard(2D)	3	768
S_2	2	trivial + sign	2	512
S_3	6	trivial + sign + standard(2D)	4	1,024

3 The Complement Split

Standard random train/test splits do not isolate symmetry-specific generalization. A model might generalize from $(a, +, b)$ to $(b, +, a)$ through embedding interpolation rather than by learning commutativity. To measure the group expert’s structural contribution, we need a split where every test example *requires* symmetry-specific transfer.

Definition. For a symmetric function f on n inputs with symmetry group G , the *complement split* assigns exactly one ordering per G -orbit to training and all remaining orderings to test. Formally, for each orbit $\mathcal{O} = \{g \cdot (a_1, \dots, a_n) \mid g \in G\}$, we select one representative $\sigma \in \mathcal{O}$ for training; the remaining $|\mathcal{O}| - 1$ elements go to test.

For S_2 on pairs $\{a, b\}$: train on $(a, +, b)$, test on $(b, +, a)$. Every test example is a reversal of a training example.

For S_3 on triples $\{a, b, c\}$: train on 1 ordering, test on the other 5. The generalization ratio is 1:5—substantially harder than S_2 ’s 1:1.

Composition split (variant): train on identity + transpositions (4 orderings), test on 3-cycles (2 orderings). This tests compositional generalization: the 3-cycle orderings are compositions of transpositions, and the irrep basis computes them by construction ($R(\sigma_1\sigma_2) = R(\sigma_1)R(\sigma_2)$), while a model without group structure must learn them independently.

In all cases, a non-symmetric control operation is split randomly, providing a baseline for the model’s general learning capacity on the same data.

4 Experiments

We present five experiments, each answering one question. All use the same base architecture: opaque embeddings (one per number, shared across positions), a concatenation layer, and residual MLP blocks, followed by the group expert layer and a scalar regression head. Unless noted, $d = 128$, 2 MLP blocks, AdamW with learning rate 3×10^{-4} , weight decay 10^{-2} , balance loss $\alpha = 0.01$.

4.1 Does complement transfer work? (S_2 , Arithmetic)

Task. $(a, \text{op}, b) \rightarrow \text{result}$, $\text{op} \in \{+, -\}$, $a, b \in [0, 20)$. Complement split on addition pairs: train on $(a, +, b)$, test on $(b, +, a)$. Subtraction split randomly.

Result. GROUP-MOE achieves 48.4% complement accuracy vs. 31.6% for the baseline—a 53% relative improvement. The effect is consistent across 3 random seeds (Table 2).

Router behavior. The router routes 95% of addition examples to the S_2 expert and 35% of subtraction examples—clean operation discrimination. Per-pair analysis shows GROUP-MOE exclusively solves $2.5\times$ more pairs than the baseline (40 vs. 16 out of 190).

Table 2: S_2 complement transfer across seeds (num_range=20).

Seed	GROUP-MoE	Baseline	Relative gain
42	48.4%	31.6%	+53%
123	45.8%	27.9%	+64%
7	35.3%	33.2%	+6%

Balance loss ablation. $\alpha = 0.01$ is the sweet spot. $\alpha = 0$ still produces complement transfer (+30% relative) but with weaker router discrimination. $\alpha = 0.1$ forces excessive routing and hurts performance.

4.2 Is it the group structure or just the routing? (S_3 , Three-Way)

A skeptic could attribute the advantage to the routing architecture rather than the group structure. We add a controlled middle ground: STANDARDMOE, which has the same router, the same project-inject dimensions ($d \rightarrow k \rightarrow d$), the same residual blending—but replaces the fixed irrep matrix $R(g)$ with a *learned* $k \times k$ matrix W . The three models have nearly identical parameter counts ($\sim 337K$).

Task. $(a, b, c, \text{op}) \rightarrow \text{result}$, $\text{op} \in \{\text{sum}, \text{nonsym}\}$, num_range=15. Two split modes: complement (1 ordering trains, 5 test) and composition (identity + transpositions train, 3-cycles test).

Table 3: Three-way comparison on S_3 ternary task (num_range=15).

Model	Complement (1→5)	Composition (4→2)
GROUP-MoE (irrep $R(g)$)	92.1%	99.0%
StandardMoE (learned W)	88.2%	98.1%
Baseline (no expert)	86.1%	96.2%

Decomposition. On the complement split, the group structure contributes +3.9pp beyond what the routing architecture alone provides (+2.1pp). On the easier composition split, the group contribution is +0.9pp vs. +1.9pp from routing. **The harder the generalization, the more the algebraic structure matters.**

4.3 Does irrep composition generalize? (Transpositions $\rightarrow$ 3-Cycles)

The strongest theoretical prediction of group representations: since $R(\sigma_1\sigma_2) = R(\sigma_1)R(\sigma_2)$, the expert should work for 3-cycles without ever training on them.

Setup. Composition split: train on identity + 3 transpositions per triple (generators of S_3), test only on the 2 three-cycle orderings (compositions).

Result. GROUP-MoE achieves 98.5–99.0% on 3-cycle orderings never seen during training. The projection P , learned from transposition examples, generalizes because the irrep subspace is shared across all group elements. At num_range=10, the router achieves perfect discrimination: 100% S_3 routing for the symmetric operation vs. 74% for the non-symmetric control.

4.4 Can the router discriminate between groups?

Nested groups ($S_2 + S_3$). With both S_2 and S_3 experts available, the router routes $\sim 70\%$ of all operations to S_3 regardless of operation type. This is *rational*: since $S_2 \subset S_3$, every S_2 transformation is also an S_3 transformation. The router discovers the subgroup relationship without supervision.

Non-nested groups ($\mathbb{Z}_2 + \mathbb{Z}_3$). With genuinely disparate groups (neither is a subgroup of the other), the router correctly dispatches $\mathbb{Z}_2$ operations to the $\mathbb{Z}_2$ expert at 76%. The non-symmetric control routes primarily to pass-through (15%) or uses the larger expert as general-purpose compute.

This confirms the ASIC analogy: the router learns which algebraic unit to invoke when the units have non-overlapping capabilities.

4.5 Does it work in a transformer?

Setup. 4-layer transformer encoder ($d = 256$, 4 heads). Each of (a, op, b, c) is a separate token with learned positional embeddings. GROUP-MoE replaces the FFN in one block.

Result. All three models (Group-MoE, StandardMoE, Baseline) reach 100% complement accuracy. Self-attention handles permutation mixing natively for short sequences, providing functional equivariance without group structure. GROUP-MoE is a zero-degradation drop-in FFN replacement, confirming architectural compatibility. The group expert’s marginal value will manifest on tasks where attention alone cannot resolve the symmetry.

5 When Does Algebraic Structure Beat Memorization?

Our experiments reveal a consistent pattern that we frame as the **lookup table (LUT) vs. algebra** tradeoff.

Memorization wins when: (1) the function decomposes into per-element contributions (e.g., $\text{sum} = \text{per-number values}$), so the effective table size is $O(n)$ not $O(n^k)$; (2) the model has enough capacity relative to the table size; (3) attention provides permutation mixing natively; (4) enough orderings are observed that interpolation suffices.

Algebraic structure helps when: (1) the complement split isolates symmetry transfer from memorization; (2) the model’s backbone does not already handle permutation mixing; (3) the effective lookup table has gaps that the group structure can fill; (4) the irrep composition property enables zero-shot generalization to unseen composed elements.

The scaling prediction. The group S_n has $n!$ elements. For S_{10} , this is 3.6 million; for S_{20} , 2.4×10^{18} . No model memorizes 10^{18} entries, but the irrep basis for S_{20} remains structured and compressible. Our synthetic experiments operate in the small-table regime where memorization is competitive. The compelling case for GROUP-MoE is the *large-table regime*—large groups, long sequences, combinatorial state spaces—where the factorial growth of the lookup table defeats memorization.

The developmental analogy is apt: humans learn multiplication tables by rote before understanding algebra. For small tables, rote memorization is strictly easier. Algebra pays off when the table is too large to memorize—and in the limit, it is the *only* strategy that works.

6 Related Work

Geometric deep learning. Cohen & Welling [2] introduced group equivariant convolutions; Bronstein et al. [1] unified CNNs, GNNs, and transformers under the equivariance framework; Weiler & Cesa [3] developed general steerable CNNs using irrep decomposition. All enforce symmetry in every layer. GROUP-MoE uses the same irrep machinery but makes it optional via routing.

Mixture of experts. Shazeer et al. [4] proposed sparse MoE with gating; Fedus et al. [5] simplified to top-1 routing with load balancing. Our router and balance loss follow this design, but our experts have algebraic structure rather than being generic MLPs. Kang et al. [6] propose “Mixture of Group Experts” using group sparse regularization for expert diversity; despite the shared name, their “groups” refer to expert grouping, not group theory.

Learned group representations. MatrixNet [7] learns matrix representations of group elements from data, achieving compositional generalization via learned generator representations. Our approach is complementary: we fix the irrep structure (guaranteeing exact composition) and learn the dispatch (routing).

Symmetry detection. Dehmamy et al. [8] discover continuous symmetries via Lie algebra convolution; Benton et al. [9] meta-learn invariances. These discover symmetries post-hoc. GROUP-MoE provides exact group structure as architectural options with learned selection.

Compositional generalization. Gordon et al. [10] hypothesize language compositionality is group-equivariance and build equivariant seq2seq models. We share the hypothesis but provide group structure as optional expert modules rather than mandatory architectural constraints.

Functional vs. structural equivariance. Walters [11] (“Ordering Is Not Invariant”) showed LLMs achieve functional equivariance without structural equivariance. GROUP-MoE is the architectural response:

providing structural equivariance as an option.

7 Discussion and Future Work

The ASIC analogy. Group experts are cognitive arithmetic units—domain-specific algebraic accelerators hardwired in the irrep basis, dispatched by a learned router. The analogy to hardware evolution is not merely metaphorical: the irrep basis provides the same structural compression that fixed-function hardware provides over general-purpose computation, and the router serves the same dispatch role as a processor’s instruction decoder.

Scaling to combinatorial regimes. Our synthetic experiments validate the mechanism but operate where memorization is competitive. The key prediction: for S_n with large n , the $n!$ growth of the lookup table will defeat memorization while the irrep basis remains tractable. Testing this requires implementing larger permutation groups or finding natural tasks with large effective symmetry groups.

Language modeling. Natural language contains implicit symmetry: entity permutations, fact reorderings, syntactic transforms. These create effective symmetry groups whose “multiplication tables” are too large for rote memorization. Inserting GROUP-MoE into language models and testing with entity permutation probes [11] is the natural next step.

Symmetry discovery. Currently the groups are specified a priori. A compelling extension: learn which groups are useful from data, by treating the irrep structure as differentiable or using a library of candidate groups with learned selection.

Approximate symmetry. Real data has approximate, not exact, symmetry. Soft irrep decomposition with learned tolerance could extend GROUP-MoE to noisy settings.

8 Conclusion

GROUP-MoE demonstrates that neural networks can learn to dispatch to algebraic fixed-function units. The complement split methodology cleanly isolates symmetry transfer from memorization. The three-way comparison shows the advantage comes from group structure specifically (+3.9pp on S_3 complement), not just routing architecture (+2.1pp). Compositional generalization confirms the theoretical payoff: irrep composition enables near-perfect zero-shot transfer to composed elements (99.0%). The router learns group-theoretic relationships—subgroup containment, non-nested discrimination—without supervision.

These are proof-of-concept results on synthetic tasks where memorization is competitive. The architecture’s value will compound in regimes where the effective symmetry group is too large to memorize—exactly the combinatorial territory where current models struggle. GROUP-MoE provides the algebraic tools; identifying the right large-scale application is future work.

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A Group Representation Details

All irrep matrices are verified to satisfy the group composition property: $R(g_1)R(g_2) = R(g_1g_2)$ for all pairs (g_1, g_2) . Full multiplication tables and irrep matrices are available in the codebase (`src/groups/representations.py`).

B Hyperparameter Sensitivity

Balance loss α . Tested at $\alpha \in \{0, 0.01, 0.1\}$ on S_2 arithmetic complement split. $\alpha = 0.01$ is the sweet spot: $\alpha = 0$ provides complement transfer without router discrimination; $\alpha = 0.1$ forces excessive routing.

Model dimension d . Tested at $d \in \{32, 64, 128, 256\}$. Results are qualitatively consistent; the group structure’s advantage is proportionally similar across scales.

Number range. At very small `num_range` (≤ 10), embeddings lack sufficient training context and all models underperform. At `num_range` ≥ 15 , the sum function is learnable and the complement split provides clear signal.

C Scaling Analysis

We conducted two scaling studies to determine whether the group advantage grows with problem size.

num_range sweep (10–50, $d = 128$ and $d = 32$): all models converge to 100% complement accuracy by `num_range` = 25. The sum function decomposes into per-number values, so the effective table size grows as $O(n)$ rather than $O(n^3)$. This means `num_range` scaling does not create the combinatorial challenge needed to separate the models.

Coverage sweep (1–4 orderings per triple, `num_range` = 15): differentiation exists only at $k = 1$ (17% coverage, 1:5 generalization). By $k = 2$ (33%), all models converge to >99%. The jump from $k = 1$ to $k = 2$ is the critical transition where even a single additional ordering provides enough context for memorization to succeed.

These results confirm that the sum function, while useful for validating the architecture, does not create a combinatorial scaling challenge. The right scaling axis is group order ($n!$ growth), not input range (n growth).